

- 8 Oxygen-15 ($^{15}_8\text{O}$) is radioactive and has a half-life of 2.04 minutes.

The decay of oxygen-15 produces positrons. For this reason, oxygen-15 is sometimes used as a tracer in positron emission tomography (PET scanning).

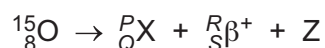
- (a) State what is meant by a tracer.

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..... [2]

- (b) The equation for the decay of oxygen-15 is



where X is the nucleus formed during the decay and Z is another particle.

- (i) State the values of the integers P , Q , R and S .

$P =$ $R =$

$Q =$ $S =$ [2]

- (ii) State the name of particle Z.

..... [1]

- (c) (i) Define the activity of a sample.

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..... [1]

- (ii) Calculate the decay constant of oxygen-15. Give a unit with your answer.

decay constant = unit [2]

